Embodied KRR for Robot Agents

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A robot that handles food has to reason about what it cannot see: how ripe an avocado is, which side its seed coat sticks to, how strong an egg's shell is and how old the egg is. We argue that knowledge representation and reasoning for such agents must be *embodied*. The symbols are grounded in a semantic digital twin of the item's body, beliefs are distributions over complete candidate items, and every decision is checked by simulating the item's mechanics before the robot acts.

We show this on two foods. On the avocado, a held-out microtheory reaches 30/30, 28/30 and 30/30 successful halvings across ripeness classes, refuses a higher-scoring knife strike as unsafe, and revises its own concepts when a request exposes a gap. On the egg, a conceptual CRAM layer turns the same ingredients into an action language derived from the ontology: belief-decided strikes crack 30 of 30 held-out eggs cleanly. All of it is MEASURED in simulation on partly ILLUSTRATIVE laws, and the failures (overconfident retries, misspecified beliefs) are reported as results.

Each twin plays two roles. For the robot, the avocado is the referent of a designator in its perception–action loop. For the researcher, it is an object of science and engineering: a population with known laws on which a category of manipulation problems is defined and studied. On that population, executives are designed and verified to be correct, complete and optimal for the category, with the evidence for each property stated (section 9).

These representations and reasoning tools are generated, not hand-written. An AI agent follows layered procedures with a gate per layer. It scopes the task, builds a sourced twin, extends the AICOR framework with a knowledge module and its theory, and produces proof obligations along with the model. Some are discharged formally, by OWL reasoning, answer-set optimality and z3 proofs of derived lab plans. The same procedure has produced twins of a machine, laboratory devices, tools, meals, a laboratory cell, an apartment kitchen and a factory site (sections 12–16).

1 Why food needs embodied knowledge

What decides whether a food task succeeds is usually hidden inside the item. In an avocado, ripeness decides whether the halves come apart cleanly or the flesh tears. Which side the seed coat sticks to decides whether the cavity is clean after pitting. In an egg, shell strength and age decide whether a strike cracks it, shatters it or does nothing, and whether the egg is still fit for the batch.

None of these can be read off a symbol such as `Ripe(avocado-1)`. The robot has to estimate them from touch, sight, weight and the outcomes of its own actions. It has to predict what a force or a torque will do to this particular item, and decide how hard it may press before it causes the damage it is trying to avoid.

This is what we mean by *embodied* KRR. The symbols the agent reasons with (`Bruised(Mesocarp)`, `TestaIn(Cavity)`, `floats ?egg`) are grounded in an executable model of the item's body. The model supplies what the symbols leave open: thresholds, forces, probabilities and time courses.

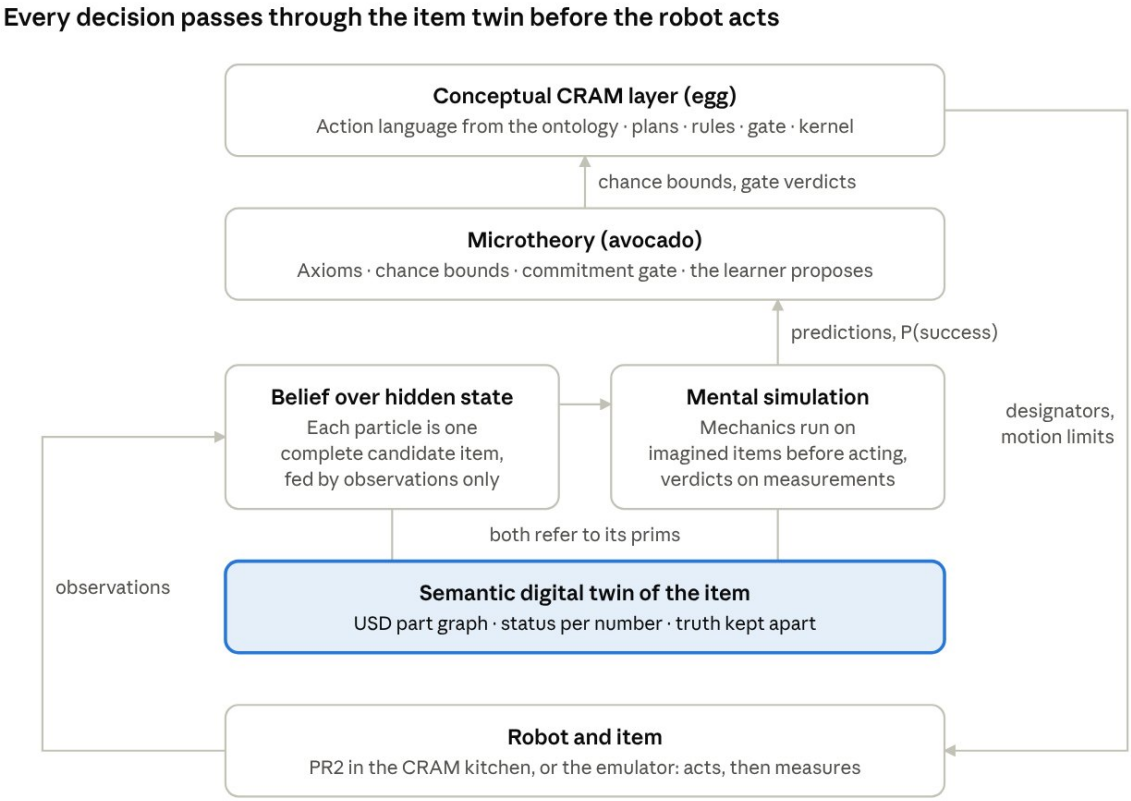
In the terms of Zhu et al., these are the ”dark” entities of the task: invisible in pixels, yet they decide its outcome (section 17).

Two foods carry the argument. They exercise different parts of the same architecture:

- **The Hass avocado** is the complete reference stack: anatomy, belief, perception, process mechanics, lifetime, a goal-directed agent on a PR2 and a held-out microtheory.
- **The egg** is the first item run through the conceptual CRAM layer. There, a research-specific action language, plan library and rule base sit on top of the twin and its belief.

The same twins serve a second purpose. Because each is a population of items with known laws, it is also an experimental object. There, the problem category itself can be studied: its corridors, its time windows, what causes its outcomes. Executives for the category can be designed and verified against it. Sections 4–8 follow the robot; section 9 follows the researcher.

2 At a glance



The robot's measurements enter only the belief. Belief and mental simulation both refer to the item's twin, and their predictions feed the microtheory and the CRAM layer, which send designators and motion limits back to the robot.

3 The semantic digital twin is the state space

Everything the robot believes, predicts or does refers to a prim of the item's part graph in USD. That makes the semantic digital twin (SDT) the state space of reasoning, not an illustration of it.

Anatomy as a part graph. The avocado is a partonomy of real tissues under `/Food/Parts`:

- the exocarp: cuticle, epidermis, hypodermis, skin parenchyma, stone-cell layer and lenticels;
- the stem button;
- the mesocarp in three colour bands with six vascular strands;
- the endocarp;
- the seed: two testa layers, two cotyledons and the embryonic axis.

Each prim carries `sdt:represents` (an IRI, so the node is kept distinct from its referent), its ontology class, `sdt:partOf`, an observability class and its volume, mass and density. The observability classes are visible, tactile, weighed, inferred and hidden. Cut states (scored, halved, pitted) are again full part graphs, so an action maps one part graph to another.

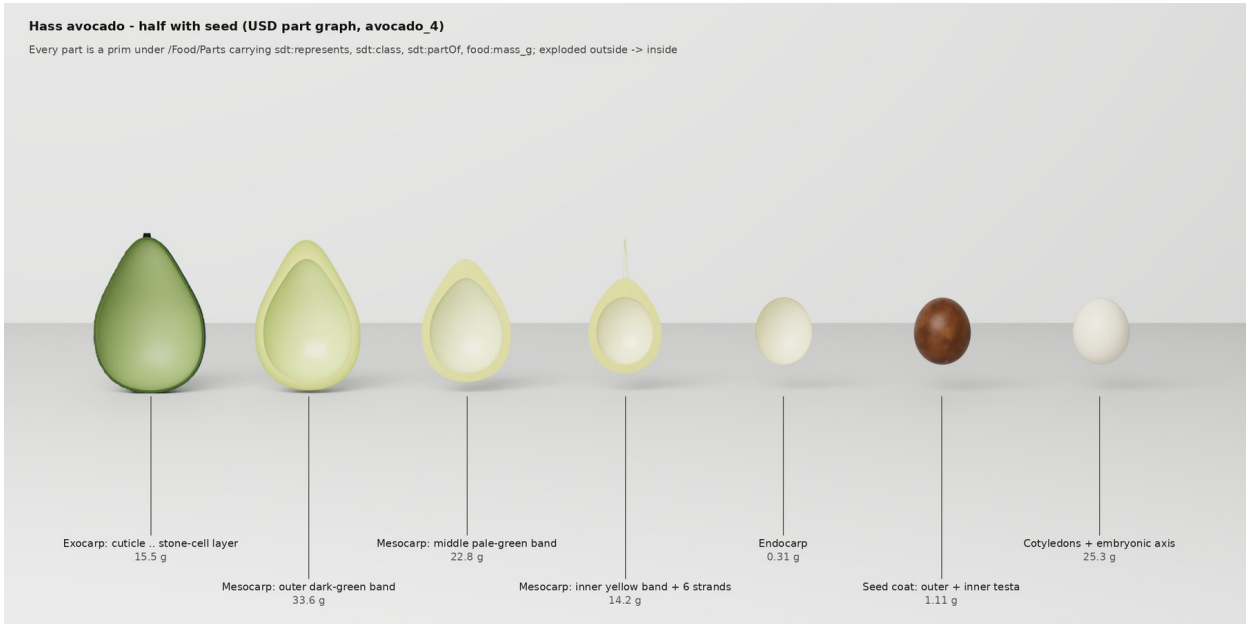


Figure 1: The avocado_4 part graph, exploded from the outside in. Every part is a USD prim under `/Food/Parts` with `sdt:represents`, `sdt:class`, `sdt:partOf` and its mass (render).

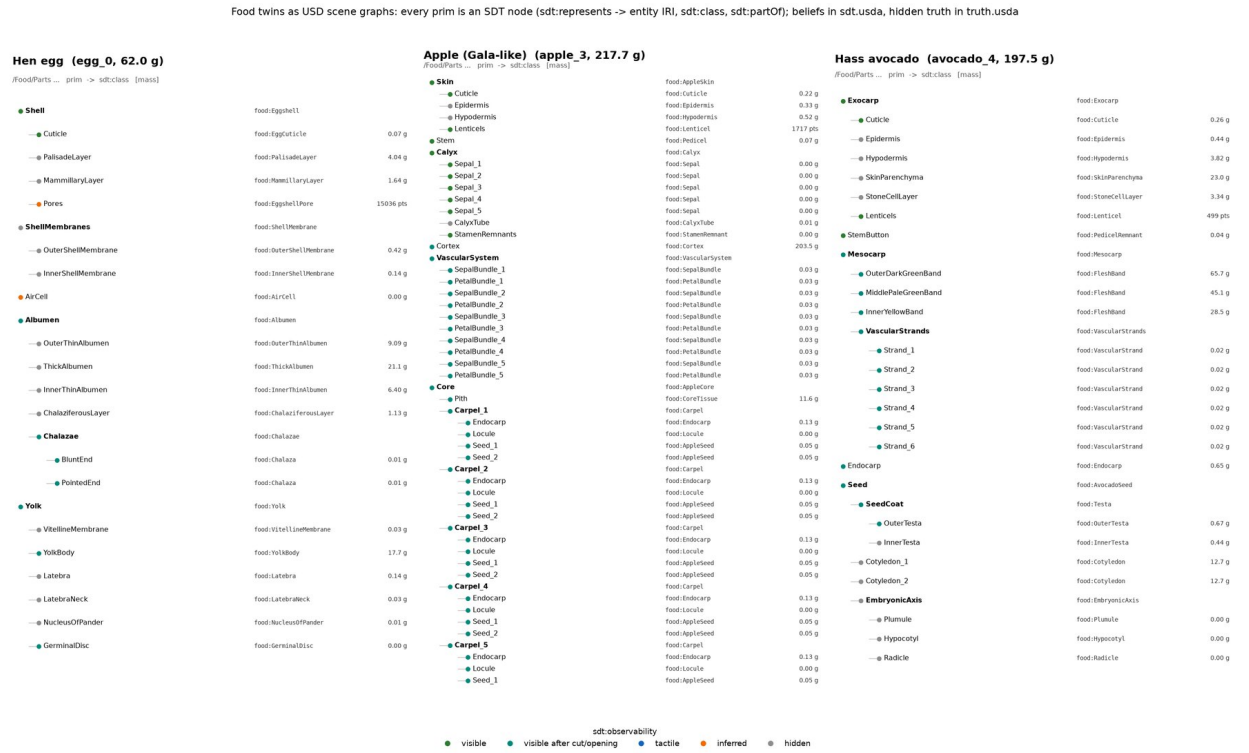


Figure 2: Three food twins as USD scene graphs. The colour of each node is its observability class: visible, visible after cutting or opening, tactile, inferred or hidden.

Numbers carry their provenance. The avocado's parameter register has 48 entries, each with a status: 14 SOURCED, 7 SOURCED-VERIFY, 2 TYPICAL, 2 DERIVED, 1 STANDARD and 22 ILLUSTRATIVE (46 %).

- The penetrometer force per ripeness stage (118 $\rightarrow$ 4.9 N) is sourced.
- Squeeze stiffness per stage comes from an Isaac FEM table (DERIVED).
- The bruise threshold, 0.6 × penetrometer force, is SOURCED-VERIFY.
- The ripeness at which the halves release cleanly, stage 2.8, is ILLUSTRATIVE.

The label travels with the number into every report.

Truth and belief are separate layers.

- `whole.usdc` holds geometry and physics.
- `sdt.usda` holds the robot's point beliefs as overs (`sdt:belief:*`, `sdt:epistemicStatus`; the seed is "unobserved (inside)").
- `sdt_belief.usda` holds the full distribution per variable.
- `truth.usda` holds the simulator's ground truth. For avocado_4 that is stage 2.958, seed 39.99 mm, testa adheres to the flesh. It is never read by the robot side; it exists to score the robot.

Actions are schema entries with guards.

- `SqueezeTest` has the guard "contact pressure below the penetration pressure of the current stage" and the effect "too hard $\rightarrow$ bruised". It observes stiffness, which updates the ripeness belief.
- `TwistApart` succeeds when the torque exceeds the flesh bridges plus the seed's adhesion. It yields halved or halved_torn, and the torque and the flesh left on the seed are themselves observations of ripeness.

The schema says *that* a guard exists. The twin says *where* it lies for this item.

4 From a detected avocado to an SDT node in the belief state

When the robot detects an avocado on the counter, it attaches an SDT node to the avocado's entity: a 4,000-particle belief over complete candidate avocados, initialised from the register. From then on, the node answers the robot's questions, sets its action parameters, judges each step and bounds its risks. In today's twin the detection itself is simulated; the node and everything after it are not.

1. Look and detect. In the PR2 kitchen run the robot parks, raises its torso, drives to the counter, looks, and performs a type-based detection:

```
perform(viz, [DetectAction(technique=DetectionTechnique.TYPES,
                           object_sem_annotation=Avocado)], context)
avocado = world.get_semantic_annotations_by_type(Avocado)[0]
node = SDT.AvocadoSDT.on_detection("avocado_1")
avocado.sdt = node
```

- **The entity** is a semantic annotation, `Avocado(Fruit)`, in the CRAM world.

- **The designator** is the perception query: ”objects of type Avocado in a region in front of the robot”. On the real robot, the same query goes to the RoboKudo perception system as `ObjectDesignator(type='avocado')`.
- **What is simulated.** In simulation the query is answered from the world model: class, identity and pose are ground truth. The only real check is that the fruit is visible in a ray-traced 256 × 256 segmentation mask from the head camera. Detection updates an entity that already exists; it does not create one.

2. Create the node. `on_detection` draws 4,000 particles from the avocado register. That covers length, diameter, neck, skin, seed size and offset, ripeness stage, ripening rate, friction and stem button, plus a testa-side flag per particle. The discrete state is set to whole, seed in, seed coat on the seed.

The prior is wide by design:

- ripeness stage 2.81 ± 1.04;
- length 98.0 ± 5.9 mm;
- diameter 65.9 ± 3.5 mm;
- mass 214 ± 26 g;
- penetrometer force 33 ± 30 N.

3. First look. A camera image of the same fruit then enters as photo observations: shape, skin colour and stem button.



Figure 3: The image the node folds in at detection: an offline path-traced render of avocado_4 on the counter, not a live head-camera frame.

One look narrows most what the skin shows and leaves what it hides untouched:

Variable	Prior sd	After photo sd	Ratio
Penetrometer force	30.1 N	5.3 N	0.18
Squeeze stiffness	1.56 N/mm	0.52 N/mm	0.34
Aspect ratio	0.119	0.047	0.39
Ripeness stage	1.04	0.53	0.51
Length	5.9 mm	4.2 mm	0.72
Mass, seed length	—	—	0.99–1.01

The ripeness mean moved from 2.81 to 3.97, against a true 2.96. The renderer darkens the skin, and the colour likelihood is not calibrated on renders. Section 6 shows how the squeeze and the torn twist pulled the belief back.

From a real photo. The same vision pipeline turns a real photograph into a node, and labels every estimate:

- **OBSERVED:** aspect, neck, colour, relief, stem button, pose;
- **ESTIMATED:** diameter, ripeness, pebble scale;
- **PRIOR:** length without a scale reference, and the whole interior.

The pipeline segments with k-means and GrabCut, fits the axis profile to the twin's meridian, and white-balances on the fruit's highlights.

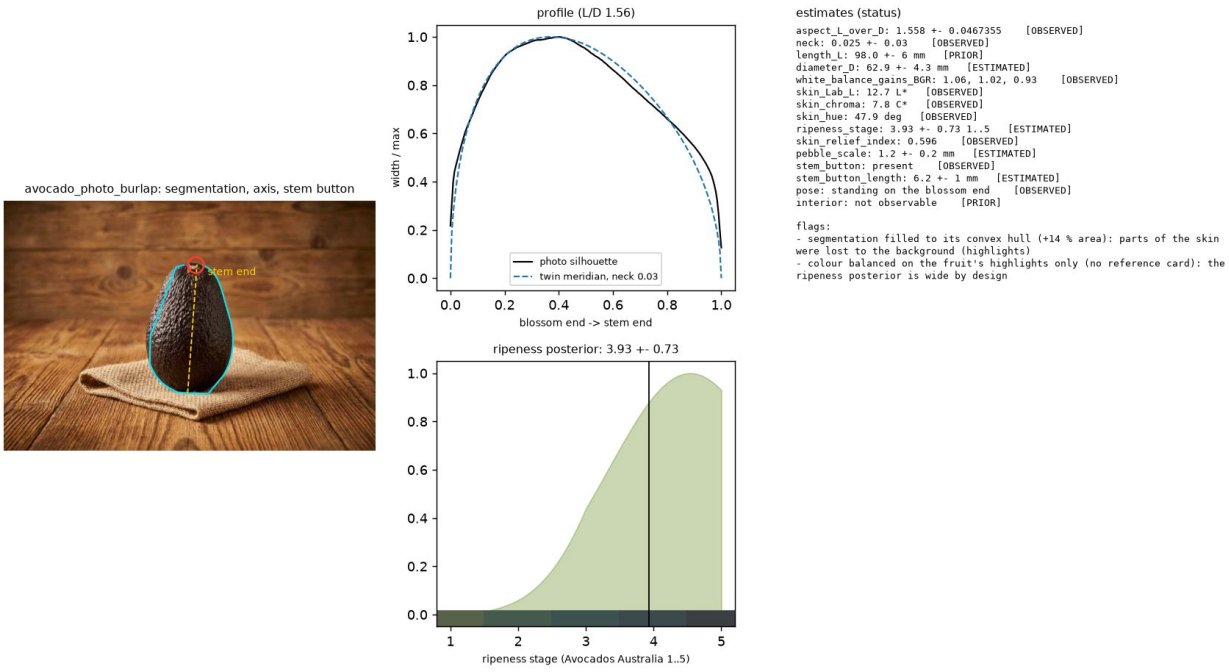


Figure 4: A real photo to a node. Ripeness 3.93 ± 0.73 (ESTIMATED), length 98 ± 6 mm (PRIOR), and the flags the pipeline raises about its own result: the mask was filled to its convex hull, and colour was balanced without a reference card.

What the node is. The node is not the fruit.

- **Separate from the body.** The node "avocado_1" is held beside the CRAM body `avocado_whole_skin`, not inside it.
- **Outlives the body.** When the fruit is cut into halves, new bodies appear, and the kitchen re-attaches the same node to each piece.

- **Ground truth elsewhere.** The truth is kept in a separate object that the robot never queries.
- **In USD.** The same structure is written as layers:
  - `sdt.usda` holds point beliefs and per-part epistemic status. The mesocarp's bruise state is "unknown (no squeeze history)"; the seed is "unobserved (inside)".
  - `sdt_belief.usda` holds, per variable, mean, sd, p05/p50/p95, a 24-bin histogram, the effective sample size and the observation history.

What the node is used for. Every number below is from the recorded PR2 run (kinematic simulation, MEASURED):

Use	How the node serves it	In the run
Answering questions	18 queries over the particles and imagined items	"How ripe?" P(ready) 0.13, ready in about 0.7 d. "Where is the seed?" Blade stops 13–19 mm at the side, 32–41 mm at the top
Choosing a force	Strongest of 4, 3, 2 N with P(bruise) < 0.02	P(bruise) at 4 N is 0.21 after the photo, so 3 N is chosen (P 0.000)
Planning the cut	Seed outline from the belief means plus a 2 mm margin	Ellipse about 22 × 19 mm around the believed seed
Planning the hold	Pads at least 10 mm clear of the blade, from the believed shape	Only a pinch at the middle qualifies: 10.0 mm clearance, 48 mm wide
Bounding motions	1.5 × the imagined 95th-percentile load	Twist 0.30 N m, seed removal 6.5 N
Monitoring	Place each measurement in the imagined distribution	Squeeze as expected; twist **unexpected** (torn), stage 3.80 → 3.00
Risk	Chance bounds over imagined items and their lifetime	Knife twist rejected (hazard 1.00); on the counter edible 0–3.8 d, P(spoiled) > 5 % from 4.5 d

The workbench queries the node's records directly, for example `an(entity(belief_variable).where(belief_variable.variable == 'ripeness_stage'))`. The researcher sees the same belief the robot acts on.

Where the loop is still open.

- **Simulated perception.** Detection reads class and pose from the world model, and the photo is a fixed render, not the head-camera frame.
- **Parameters reported but not applied.** The motion limits are reported but not yet passed to execution. The 3 N squeeze is always safe, because the penetrometer model never falls below about 6 N.
- **Node not persisted.** The CRAM node is not written back to its USD layer during the run.
- **Belief overconfident.** At the end, the seed-length band (40.0–42.8 mm) excludes the true 39.99 mm.

5 Belief: one particle is one complete candidate item

The robot's knowledge of an item's hidden state is a particle belief in which each particle is a whole candidate item, not a value of one variable. Derived quantities are computed per particle, so their dependencies carry information across variables.

Avocado. The SDT node holds 4,000 particles.

- Each samples length, diameter, neck, skin thickness, seed length, width and offset, ripeness stage, ripening rate, skin friction and stem button, plus a bruise flag and a testa-side flag.
- Mass, penetrometer force, Young's modulus and squeeze stiffness are computed from these per particle.
- Weighing therefore narrows size, and squeezing narrows ripeness and the colour it implies.
- Time prediction uses each particle's own ripening rate.

Egg. The adapter holds 2,000 particles of a full-egg model.

- Each samples shell thickness, shell modulus, equatorial and axial breaking force, friction, yolk fraction, laying density, fresh Haugh unit and yolk index, an ageing rate and the age in days, plus a micro-crack flag (prior 0.03, ILLUSTRATIVE).
- From these come mass, specific gravity, air cell, float posture, a direction-dependent crack force and the force at which the egg slips from the grip.

Observations are the only way in. Every update is an `Observation(kind, value, t)` through a measurement model that sits next to the belief and has its own noise and status.

The avocado has twelve such channels:

- calliper (1 mm) and scale (2 g or 3 %);
- photo shape and skin colour;
- stem button;
- squeeze stiffness, with 20 % model error;
- seed length and blade stop depths (1.5 mm);
- twist outcome and torque;
- spoon force;
- seeing the seed coat;
- seeing a bruise.

The egg adds the carton date, candling, the float test (tilt, 8° sd), grip, strike outcome and the opened contents.

Hearing the item's own answer. Outcomes of actions are observations too. A twist that tears the flesh says the avocado was less ripe than believed. A strike that does not crack the egg says its shell is stronger.

On the egg this needed care. The first strike likelihood was too sharp: after one no-crack, the next strike looked certain and shattered eggs. The strike likelihood is now tempered, `lik = 0.9 · lik + 0.1/4` (ILLUSTRATIVE).

Convergence on a known instance. For avocado_4, MEASURED on the simulator with ILLUSTRATIVE noise:

Step	Ripeness belief	Truth
Prior	2.75 ± 1.04	2.96
Photo colour	2.99 ± 0.80	2.96
Photo at 48 h	4.07 ± 0.65	4.11
Squeeze at 72 h	4.66 ± 0.38	4.68

Step	Ripeness belief	Truth
Cut at 96 h	4.89 ± 0.21	5.00

The seed length estimate after the cut was 39.15 ± 1.27 mm against 39.99 mm.

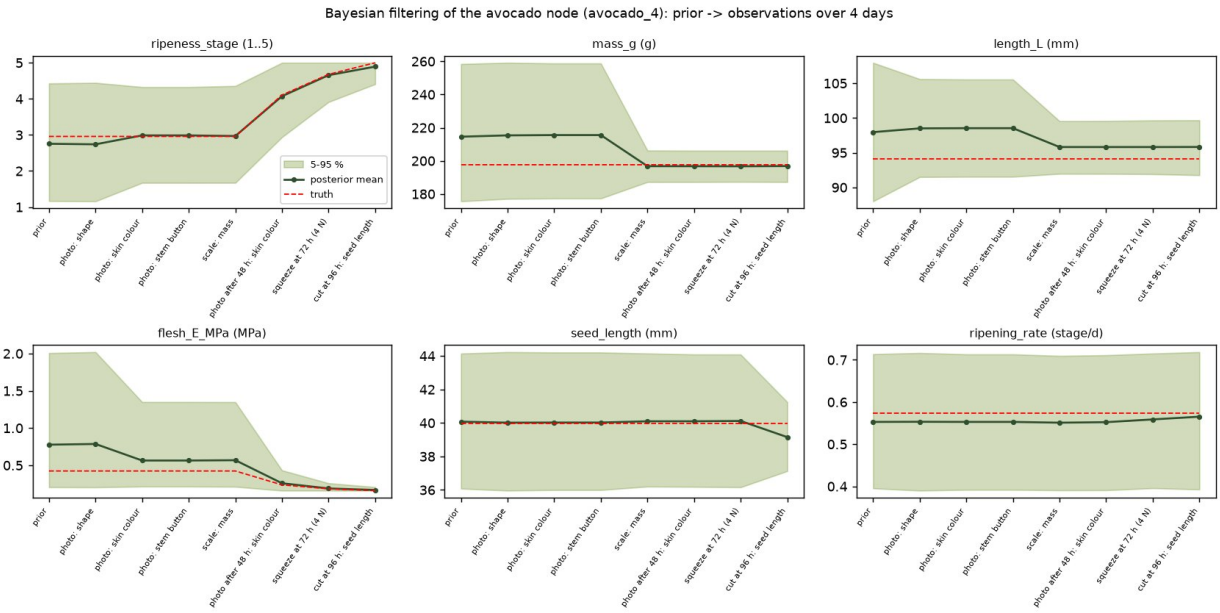


Figure 5: Filtering the avocado_4 node over four days. Each observation narrows the variables it informs: the scale narrows mass and length, the squeeze narrows the flesh modulus, the cut narrows the seed. The ripening rate stays wide because nothing observes it directly (MEASURED in simulation, ILLUSTRATIVE noise).

For the egg, on 24 held-out instances, the truth lay inside the 90 % band in 99 % of steps for age, 94 % for crack force and 86 % for yolk index (MEASURED on the emulator).

6 Mental simulation and execution monitoring

Before acting, the robot runs the item's process mechanics on items drawn from its belief: imagined avocados and imagined eggs. Prediction is simulation of the body, not a lookup table, and it yields both what to expect and how to bound the motion.

This is the robot's version of the "game engine in the head" that Battaglia, Hamrick and Tenenbaum proposed for human physical judgment: approximate, probabilistic simulation over an uncertain scene. Here it is used to act as well as to predict (section 17).

Imagined items. For the avocado, the halving and pitting mechanics run on 150 particles in about 1 ms of reduced-order geometry each. Motion limits are set at 1.5 × the imagined 95th-percentile load.

In the PR2 run that gave a twist limit of 0.30 N m (imagined p95 0.20 N m) and a seed-removal limit of 6.5 N (imagined p95 4.3 N). For the egg, the strike model gives P(intact, check, cracked open, shattered) per particle from the ratio of impact force to crack force, integrated over 10 % force noise.

Monitoring places the measurement in the imagined distribution. For each step:

- p_out = the imagined share of the observed discrete outcome;

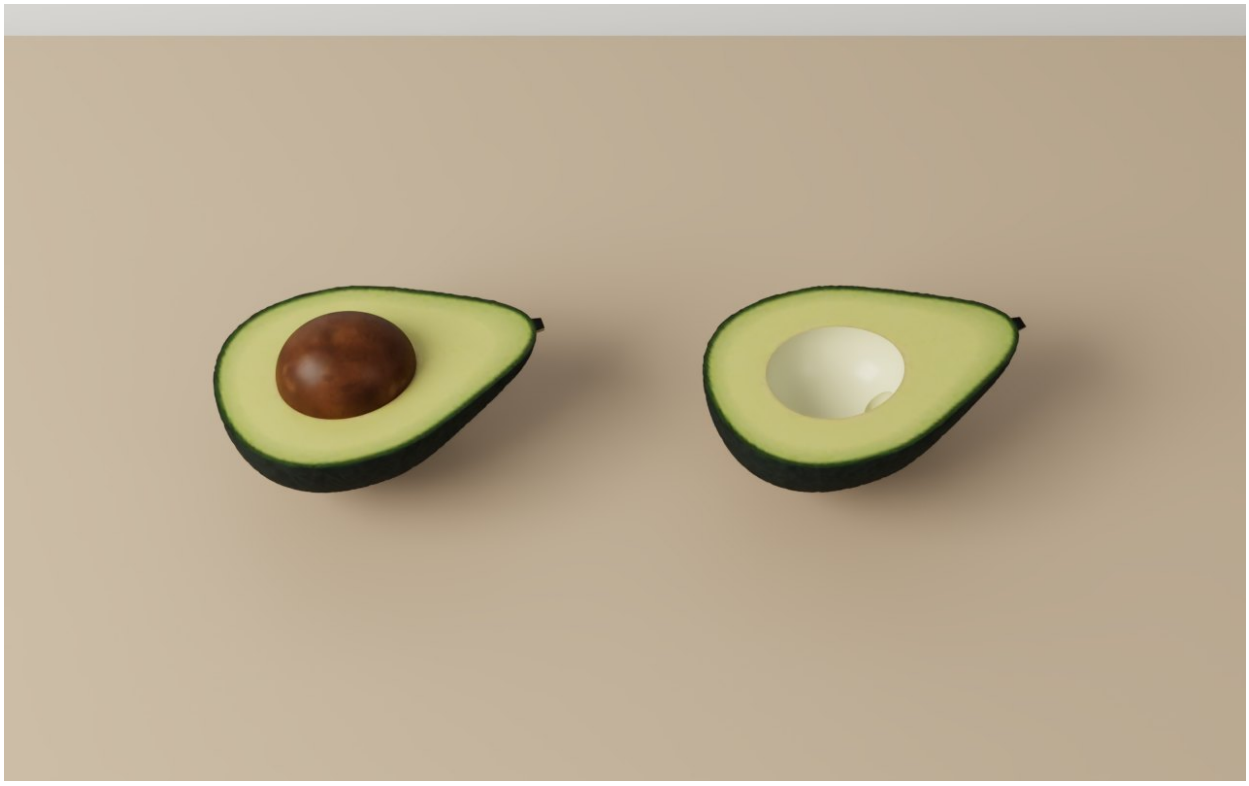
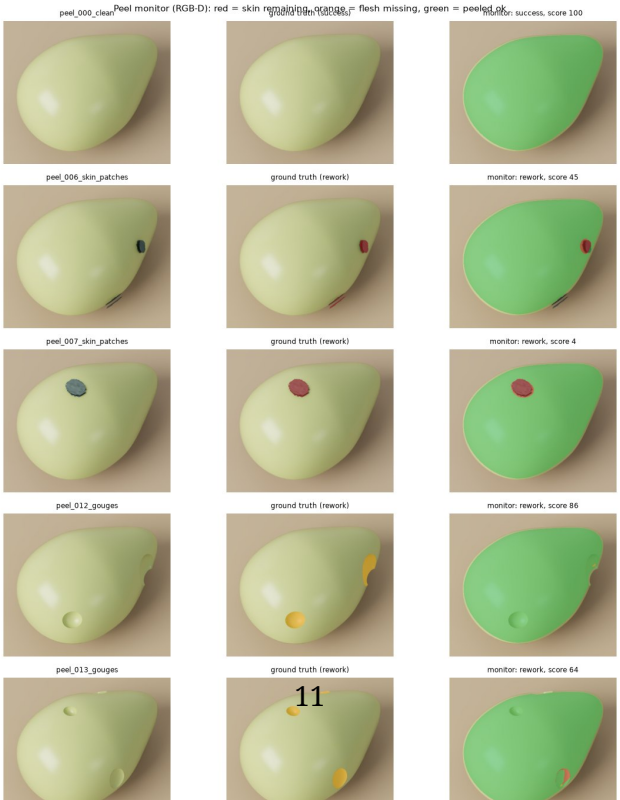


Figure 6: A process state written back as USD: the ripe scenario after TwistApart, rendered from the emulator's part graph. The same state is what the robot imagines, drawn from its belief.

- q = the quantile of the measured force or torque.

Verdict	Condition
surprising	p_out < 0.05, or q outside [0.025, 0.975]
unexpected	p_out < 0.2
as expected	otherwise

Only the measured part then enters the belief. The monitor never sees the truth.



Step	Imagined	Measured	Verdict	Ripeness belief after
Squeeze 3 N	2.05–3.97 N/mm	3.40 N/mm	as expected	3.79 ± 0.52
Cut round the seed	P(scored) 1.00; 6.2–14.7 N	scored, 13.10 N	as expected	3.80 ± 0.53
Twist apart	P(torn) 0.07; 0.07–0.15 N m	torn, 0.18 N m	**unexpected**	3.00 ± 0.38
Remove seed	P(testa left) 0.49; 2.7–5.7 N	testa left, 3.37 N	as expected	3.16 ± 0.35

The truth was 2.96. The rendered photo had suggested 3.97, because the renderer darkens the skin. The squeeze is a weak cue, with 20 % model error. The torn flesh was the observation that corrected the belief: the robot learned the fruit was less ripe from the outcome it had not expected.

After pitting, P(testa adheres to flesh) was 0.96, and the seed length estimate was 41.45 ± 0.85 mm against 40.0 mm.

7 Decisions: chance bounds, value of information and safety

The agent decides under its belief with chance-constrained choices. Goals must hold with at least a bound, prevented effects may occur with at most a bound, and safety rules have bound 0 and are never traded for value.

Bounds are explicit objects. The avocado agent states each as `Bound(effect, p, hard)`.

- Goals: halves apart ≥ 0.95, seed out ≥ 0.95, clean cavity ≥ 0.9, edible at service ≥ 0.7, fresh at service ≥ 0.8.
- Prevented: bruise ≤ 0.02, spoiled ≤ 0.05, browned ≤ 0.2, seed coat left ≤ 0.1.
- Hard: hazard ≤ 0 and blade near the hand ≤ 0.

The bounds are ILLUSTRATIVE. Each plan option is simulated on 80 imagined avocados, including the agent's own repairs and the lifetime model. The agent picks the feasible plan with the best expected quality. When nothing is feasible, it takes the plan with the fewest shortfalls that breaks no hard rule, and says so. When every plan breaks a safety rule, it refuses.

Safety outranks utility. For "guacamole now", with ripeness believed at 3.97 ± 0.53:

Plan	Expected utility	Decision
Knife twist + peel	0.80	infeasible: hazard 1.00 > 0 (safety rule)
Spoon + peel	0.745	**chosen**: clean cavity 0.99, edible 0.81
Spoon, no peel	0.65	infeasible: clean cavity 0.50 < 0.90

The best-scoring plan is rejected because it puts a blade near a hand.

Time is part of the decision. For "half now, half tomorrow", the kept half is served after 15 h. Wrap, lemon or nothing all predict P(browned) = 1.00 and fail the bound. Wrap + fridge, lemon + wrap + fridge and vacuum + fridge are feasible, and vacuum + fridge is chosen at quality 0.68.

Value of information before a risky probe. A squeeze tells the agent about ripeness but can bruise.

1. The agent takes the strongest of 4, 3 or 2 N whose P(bruise) under the belief is below 0.02. In the PR2 run that was 3 N, with P(bruise) 0.000.

2. It runs a preposterior analysis: 120 imagined avocados × 120 imagined readings. The sensor's relative noise grows as the force falls.
3. Without the squeeze, it would store the fruit two days (expected utility 0.77 vs 0.68 for now).
4. The squeeze changes that choice in 8 % of imagined readings and is worth +0.012. That is above the 0.005 threshold (ILLUSTRATIVE), so it squeezes.

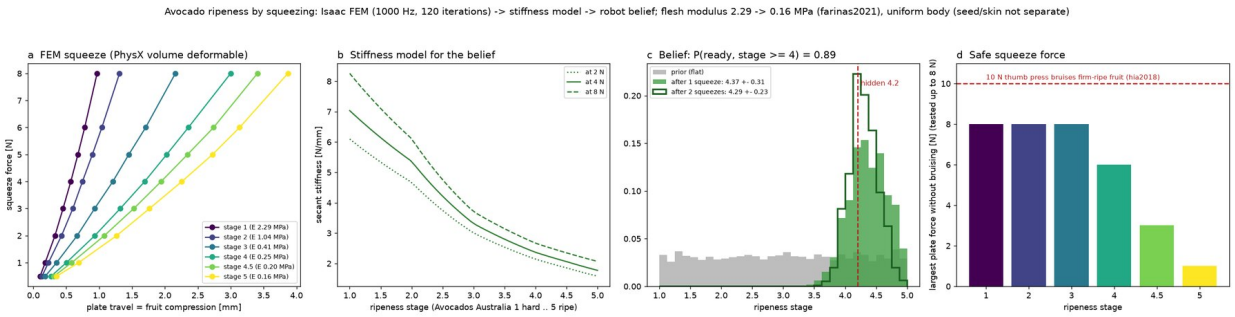


Figure 8: Why the squeeze force must come from the belief. Isaac FEM gives the force–compression curve per ripeness stage; the stiffness model turns a reading into a belief update. The largest bruise-free force falls from 8 N for hard fruit to 1 N for ripe fruit, while a 10 N thumb press bruises firm-ripe fruit.

The egg strike, decided from the belief.

- The strike surface and speed are chosen as the maximum P(crack) over the bowl rim and the counter and over 0.05–0.89 m/s, subject to P(shatter) $\leq$ 0.05 (ILLUSTRATIVE).
- Before deciding, a value-of-information probe imagines 12 scale readings. Weighing is performed only if it raises the expected P(crack) by at least 0.02.
- In the weak-shell scenario the probe fired: P(crack) 0.865 $\rightarrow$ 0.890, VOI +0.026.
- After every strike the decision is withdrawn and re-made on the updated belief.

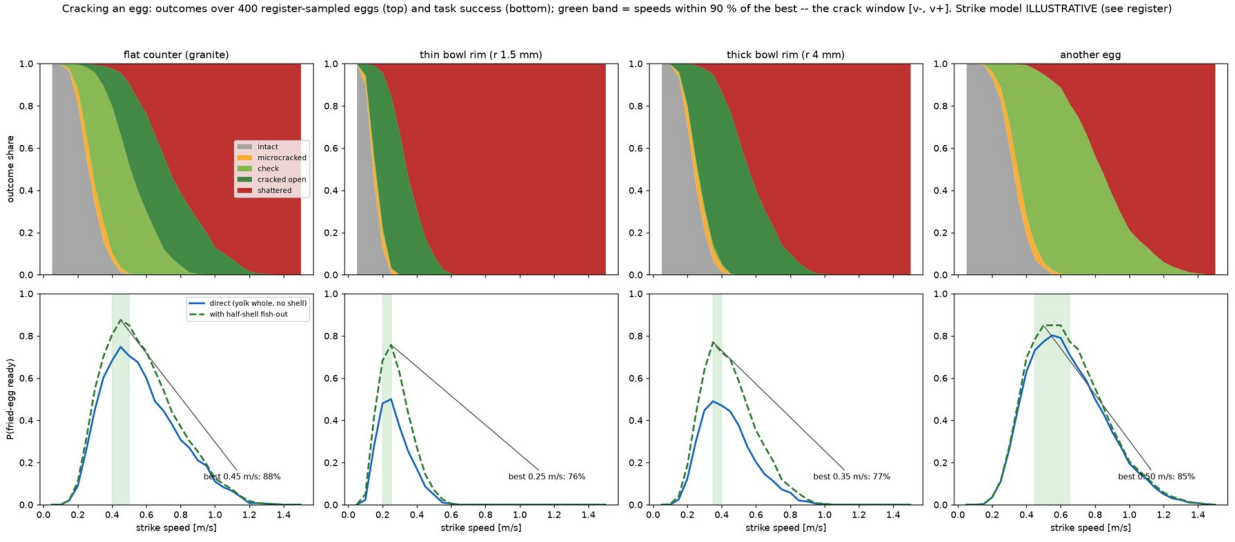


Figure 9: The egg's crack window per strike surface over 400 register-sampled eggs. Outcomes run from intact to shattered as speed rises, and the best speed differs by surface: counter 0.45 m/s (88 %), thin rim 0.25 m/s (76 %). The strike model is ILLUSTRATIVE.

The cautious policy. A robust variant also requires the first strike to keep P(shatter) ≤ 0.05 under a pessimistic copy of the belief with breaking forces × 0.6. Its likely no-crack is then treated as informative rather than as a failure.

8 The microtheory: axioms that are tested, revised and kept in charge

The avocado microtheory is a versioned theory T = ⟨Σ, A, M, C, E⟩: concepts, axioms, models, claims and evidence. Its claims are checked on held-out avocados, it is revised when a request exposes a gap, and it keeps authority over any learned policy.

Axioms in an interval language. Axioms are stated over holds(f, i), occurs(a, j) and meets(i, j). Each has a precondition, guarded add and remove branches, and a frame rule:

- **Squeeze:** `Bruised(Mesocarp)` iff the force exceeds the bruise force of the current stage.
- **TwistApart:** gives `TornFlesh` unless the flesh releases at the seed. It gives `Stuck` when the flesh bridges plus the seed's adhesion exceed the torque limit.
- **RemoveSeed:** gives `TestaIn(Cavity)` when the testa adheres to the flesh. It gives `UnsafeToolUse` when the method is a knife strike.

The requirements are phase-indexed:

- **Task A, assess:** the fruit is never bruised, and the ready decision is right.
- **Task B, halve and pit:** apart, seed out, no torn flesh, no seed coat left, not bruised, no unsafe tool use, eating quality ≥ 0.6, not spoiled.

A concept counts only if it is perceivable, actionable and traceable.

Held-out claims. Seeds 1000–1029 give 30 avocados per ripeness class and 3,120 episodes, all MEASURED in simulation.

- **Task B:** the belief-band policy succeeds 30/30, 28/30 and 30/30 for unripe, breaking and ripe fruit. Cutting immediately succeeds 0/30, 10/30 and 30/30 (McNemar 48:0, p = 7.1e-15).
- **Task A:** the closed-loop squeeze beats a 10 N thumb press 46:0 (p = 2.8e-14). The thumb bruises 24 breaking and all 30 ripe fruits. It beats photo-only 14:2 (p = 0.004).

Ablations name what matters.

- Removing the seed-coat-side concept drops Task B to 39/90 (49:0, p = 3.6e-15).
- Removing the repair, the per-fruit ripening rate or the graded ripeness changes nothing detectable. That is reported as a result.

The axiom audit. We replayed the axioms against 1,350 simulated episodes, 8,699 steps. There were 0 precondition violations, 0 branch mismatches, 0 guard mismatches and 0 frame violations.

There were also 22 counterexamples: the same abstract state and action led to different branches. Ten were seed-coat peels and seven seed removals. The relational state does not fix the outcome; the hidden testa side and stage do. The symbolic theory needs set-valued transitions or refinement by hidden attributes, and the twin shows exactly where.

Learning as theory revision. Each new request that exposed a gap produced a patch and a new held-out claim.

Version	Request → gap	Before	Patch	Held-out after
T1	firm fruit → separation depends on ripeness	10/60	admit RipenessStage, TornFlesh	88/90
T2	clean cavity → seed coat unrepresented	39/90	admit SeedCoatSide	88/90
T3	new camera → colour cue misspecified	27/30	add TactileRipeness; fitted bias +10.44 (true +10)	30/30
T4	keep a half → browning unrepresented	0/90	admit Browning	88/90
T5	sticky cultivar → outside the family	41/60	restrict scope; claim **retracted**	41/60 refuted

A commitment gate turns this into the agent's answer to "should I do it?". It answers warranted, uncertain (photo only), unsupported (knife twist: hazard P 1.00), counterevidenced (the sticky cultivar) or out-of-scope.

The theory owns authority, the learner proposes. We generated a reinforcement-learning program from the theory's own concepts: features from Σ, actions from the designators, reward from the claim's criteria. With a shield derived from the forbidden effects:

- RL from T3 matches the hand-written belief band: 30/30, 28/30, 30/30, with 0 hazards.
- RL from T1, which lacks the seed-coat concept, reaches only 12/30, 13/30, 13/30. What the theory cannot represent, the learner cannot learn.
- With the shield replaced by a soft penalty of 0.05, the learner chooses the knife strike in 90 of 90 held-out episodes and succeeds 0/90, while its training reward still rises to 0.972.

All numbers in this section are MEASURED in simulation on plant laws that are partly ILLUSTRATIVE.

9 The item as an object of science and engineering

Every twin has a second role. For the robot, an avocado is the referent of a designator: one item to perceive, believe about and act on. For the researcher, it is an object of science and engineering. It is a population of items with known laws, on which a whole category of manipulation problems can be posed, studied and solved. That second role is what lets us design manipulation executives that are correct, complete and optimal for a category of problems, and state the evidence for each claim.

	In the perception–action loop	As a science and engineering object
What the item is	One avocado on the counter, the referent of `an avocado`	A population of avocados drawn from the register, with the laws of the plant
What is uncertain	This item's ripeness, seed and testa side	How outcomes vary across the population and the scope
What is computed	A belief, a decision, a force limit for this item	Corridors, windows and causes for the category; a policy for all of it
What counts as evidence	Monitoring verdicts on this execution	Held-out verification, ablations, misspecified plants, proofs

	In the perception–action loop	As a science and engineering object
Who uses it	The robot, at run time	The researcher designing and verifying the executive

A category of manipulation problems. The microtheory defines a scoped manipulation problem as

$$\mathcal{P} = \langle R, \tau, \Psi, L \rangle, \qquad \pi \text{ meets } q \iff P_{\beta\sim\tau,\, \theta\sim L}\big[\Psi(\text{trace}_\pi)\big] \geq q$$

Here R are the roles, τ their types with parameter distributions (the register's population), Ψ a success predicate over SDT traces, and L the plant's laws. A category is a family of such problems: "halve and pit" across unripe, breaking and ripe fruit, or "crack into the batch" across fresh and stale eggs and strike surfaces. Against this definition the three properties of an executive become checkable:

- **Correct:** on every scope of the category, the executive meets q.
- **Complete:** whenever some plan in the policy language meets q, the executive finds one. When none does, it refuses, with the reason.
- **Optimal:** among the plans that meet q, the executive picks the best by the stated criterion (expected quality, least cost, highest information gain), subject to hard safety rules.

Scientific understanding: what the twin reveals about the category. Because the population is known, the twin answers questions about the problem itself, not only about one policy:

- **Corridors.** For ripe avocados the squeeze corridor is empty. Informing the ripeness belief needs at least 3.33 N, while P(bruise) is already 0.18 at 4 N and 0.61 at 5 N. So every squeeze policy leaves ripe fruit undecided (23/30 at best). That is a property of the problem, not a failure of a policy. For the egg, the grip corridor where slip and rupture both stay below 0.05 is computed per egg (0.9–19.6 N in the stale-egg run). The crack window per strike surface is mapped over 400 eggs (section 7).
- **Windows in time.** Sweeping the cut day shows unripe fruit succeeding 0/30 on day 0 and 30/30 by day 5, and ripe fruit 30/30 on days 0–2 but 2/30 on day 5. The belief-band policy is designed to land inside each class's window.
- **Causes.** Hypothesis tests on the plant separate what controls the outcome. Ripeness controls clean release: P(clean) rises from 0.09 to 0.97. Twist torque does not: torn stays at 10/30 at every torque limit. Ablations name the concepts an executive cannot do without; without the seed-coat side, success falls to 39/90.

Engineering design: what the twin lets us build and verify.

- **Correctness by statistical verification against typed scopes.** A plan is *verified* on a scope if the 95 % Wilson interval lies at or above q = 0.8, *refuted* if it lies below, *undecided* otherwise.
  - The belief band is verified on unripe and ripe fruit and undecided on breaking fruit (28/30).
  - Cutting now is refuted on unripe fruit (0/30) and verified on ripe fruit.
  - The firm 10 N thumb test is refuted on breaking and ripe fruit.
  - In the lab the same idea becomes proof: z3 shows that the derived plans are executable, safe, on time and informative enough, for every unknown initial state (section 15).

- **Completeness, conditionally.** The microtheory states a conditional proposition. Suppose:

  1. the abstract initial relations represent each physical initial state;
  2. the guards of the abstract branches cover exactly the simulator's successors;
  3. the abstract observations preserve what the policies branch on;
  4. trace extraction and goal evaluation agree.

  Then abstract and simulated goal satisfaction agree for every case, and a complete search of a finite policy language finds a solution if and only if one exists. The replay audit checks these conditions: 0 violations of preconditions, branches, guards and frames over 8,699 steps. But the 22 counterexamples break condition 2, so the avocado proposition is not yet discharged. In the lab, the planner's refusals of three requirements are proofs that no plan in the candidate frame meets them.

- **Optimality, relative to the model.**

  – The avocado agent picks the best expected quality among plans that meet every bound.
  – The egg strike is the maximum P(crack) subject to P(shatter) ≤ 0.05.
  – The lab experiment is proven least-cost, then highest-gain, within 45 candidates.

  Optimality is only as good as the model of the population. When the robot's belief overrates the shells, every first strike claims P(shatter) ≤ 0.05, yet 9 of 30 eggs shatter (section 11). Robustness to misspecification is therefore part of the design brief: the cautious policy reduced those shatters to 3, at a cost under nominal conditions.

- **The design space itself.** The theory's concepts define which executives can be expressed at all. A learner given the theory's features and a shield reaches the hand-designed executive (30/30, 28/30, 30/30). Given an older theory without the seed-coat concept, it cannot (12/30, 13/30, 13/30). Given a soft penalty instead of the shield, it buys the knife strike in every episode.

The same two roles hold beyond food. The Cell 01 culture model is both what the robot reasons about and the population on which experiment plans are proven. The factory's product model is both the robot's world and the object on which station sequences, crane capacities and collision-freedom are verified (section 16). In each case, the twin is where the category is defined, the executive designed, and the claims of correctness, completeness and optimality are made, with the status of each stated.

10 The conceptual CRAM layer: a research-specific action language over the twin

The egg runs through a conceptual CRAM layer. For one research activity it gives the researcher four things over the same twin and belief:

- an action-designator language;
- a plan library;
- a rule base;
- a kernel that performs designators.

Its point is that every keyword, goal and failure is an ontology term, so the agent can reason about its own plans with the same knowledge it uses about the world.

The language has no words of its own. The lexicon is derived from the ontology: DUL, SOMA and an egg module. The grammar comes from the task axioms: `Cracking` ⊑ `soma:Manipulating` requires a patient and a tool and admits a strike speed and an effector.

The linter reports any library construct the ontology does not entail. Today the egg activity lints clean, with 0 predicates missing from the ontology. All 8 action schemas have a goal, and every goal in its plans can be refined into an action.

Goals are templates of state evolutions. An action schema states what must become true and what must not happen over its interval:

```
(def-action-schema cracking
  (:roles (object (an egg)) (tool (a strike-surface)) (speed (some strike-speed)))
  (:pre   (held ?object))
  (:telos (:over ?i) (achieve (shell-state ?object cracked) :at (end ?i))
                     (prevent (shell-state ?object shattered) :during ?i))
  (:failures (no-crack NoCrack) (egg-shattered EggShattered)))
```

`perform` infers the goal from `:telos`. `achieve` refines a goal into a designator. Determiners mark use: `an egg` is a concept to be bound, `the egg` a referent.

Rules carry state-dependent knowledge.

```
(def-rule floats
  (:if (float-observed ?egg floats ?tilt))
  (:then (float-class ?egg floats) (stale ?egg)))
(def-rule unfit-contents
  (:if (contents-in ?egg ?cup) (stale ?egg))
  (:then (unfit ?cup)))
```

Forward chaining feeds the emptying schema's gate, `(prevent (emptied-into ?cup ?bowl) :while (unfit ?cup))`. Opening an egg directly into the batch bowl is defined in OWL as a closing event, and HermiT classifies it as a prohibited action. Closing events are refused by the gate, never handled.

Bindings carry their warrant. Role fillers are found by a climb through servants, each recording a status: recorded, then anchored, then commonsense, then assumed.

- The strike surface and speed come from the belief, anchored and revisable.
- The grip force is the geometric mean of the belief's slip/rupture corridor, also anchored.
- A 6 N default grip would be commonsense.

The kernel's cycle per action.

1. Climb the roles.
2. Instantiate the goal.
3. Skip if it already holds.
4. Check the gate.
5. Achieve location preconditions by refinement.
6. Offer a probe if its value of information is high enough.
7. Assess the warrant and a pre-act verdict with P(success).
8. Pre-empt a predicted failure that has a handler.
9. Execute.
10. Check the state trajectory against the goal.

11. Write an episode node.

Failures are typed descriptions, signalled and caught by the enclosing `with-failure-handling` by ontology subsumption. `(continue)` resumes, and other handlers unwind.

A failure can declare which goal parts a resuming handler accepts. `yolk-broken :violates ( yolk-whole ?e)` lets a batter take a broken yolk: the step reads ”goal achieved (accepted: yolk-broken)”. This facet is provisional.

One episode: a stale second egg. The plan is `two-eggs-into-batch`, run on the emulator; egg 2 has been kept 84 days at room temperature.

1. Egg 1 sinks in the float test, tilted 21°. It is gripped, struck, opened, inspected and emptied into the batch bowl.
2. Egg 2 floats upright. The rules derive `(stale egg-2)`.
3. Grip at 4.19 N: anchored, ”corridor 0.9–19.6 N, P(slip) 0.000, P(rupture) 0.000”. The warrant is rational; act now.
4. Strike on the counter at 0.42 m/s: anchored, ”max P(crack) 0.86 with P(shatter) 0.047 $\leq$ 0.05 (2,000 particles)”. It cracks.
5. Open into the cup, inspect: 0 shards. Cleaning is skipped because the goal already holds.
6. The rule `unfit-contents` derives `(unfit cup-1)`.
7. Emptying is **refused by the gate**: `(prevent (emptied-into cup-1 batch-bowl) :while ( unfit cup-1))`. The batch keeps egg 1 only.



Figure 10: What `opening` does in the physics twin: shell fragments from the crack model, 7,703 albumen particles and a rigid yolk. After 4 s, 99.9 % of the albumen is in the bowl (Isaac Sim).

A second scenario shows the pre-act verdict. A researcher forces a timid 0.12 m/s strike. The role is assumed, so the warrant is only plausible, and the prediction is P(success) 0.01. The kernel pre-empts the step (”repair-first: predicted no-crack”), re-decides the strike to 0.43 m/s on the counter with P(success) 0.88, and the egg cracks.

The same language files run in two twins:

- the emulator, which executes callables;
- a PR2 kitchen in CRAM, which puts real motion plans around the emulator’s outcomes.

Both run in a research workbench with a CRAMERA 3D view.

11 Evidence: the egg strike evaluation

On 30 held-out eggs, the belief-decided strike cracked all 30 cleanly, against 26 for the best fixed strike from the crack study, and its first-strike P(success) was calibrated (0.86 predicted, 0.87 observed). It degrades under a misspecified belief, and its retries are overconfident.

The avocado's results are given in sections 4–9. This section adds the egg's.

Setup.

- Register instances 1000–1029, one per egg.
- Emulator seed = instance, so all conditions see the same 30 eggs and the comparisons are paired.
- Plan `crack-egg-cleanly`, at most 3 strike attempts.
- To test robustness, the robot's belief is misspecified, with shell breaking forces × 0.6 (weak) or × 1.6 (strong) relative to the eggs' truth.

All results are MEASURED on the emulator; P_SHATTER_MAX, CAUTION, EPS_STRIKE and VOI_MIN are ILLUSTRATIVE.

Condition	Cracked cleanly	Shattered	Retries exhausted	Strikes per egg	Weighing probes
Belief	**30/30**	0	0	1.13	3
Fixed best strike (rim, 0.25 m/s, +15 % per retry)	26/30	3	0	1.13	0
Belief, weak	28/30	0	2	2.00	14
Belief, strong	21/30	**9**	0	1.00	30
Cautious	28/30	2	0	1.70	2
Cautious, weak	24/30	0	6	2.43	2
Cautious, strong	27/30	3	0	1.17	29

The fixed strike also left one yolk broken, with one shell shard in the cup.

Paired comparisons (discordant pairs, exact McNemar):

- Belief vs fixed strike: 4:0, p = 0.125. The direction is consistent, but n = 30 does not establish it.
- Belief vs belief-strong: 9:0, p = 0.004. A belief that overrates the shells breaks the chance bound: every first strike claimed P(shatter) $\leq$ 0.05, yet 9 of 30 eggs shattered.
- Cautious vs belief, under the strong misspecification: 9:3, p = 0.15. Caution cut shatters from 9 to 3, but not significantly. Under nominal conditions it cost 2 shatters and 0.6 extra strikes per egg; under the weak misspecification, 6 eggs ran out of attempts.

Calibration of P(success) (predicted mean $\rightarrow$ observed rate):

	Predicted	Observed	n
First strike, belief	0.86	0.87	30
First strike, six belief-based conditions	0.61	0.50	180

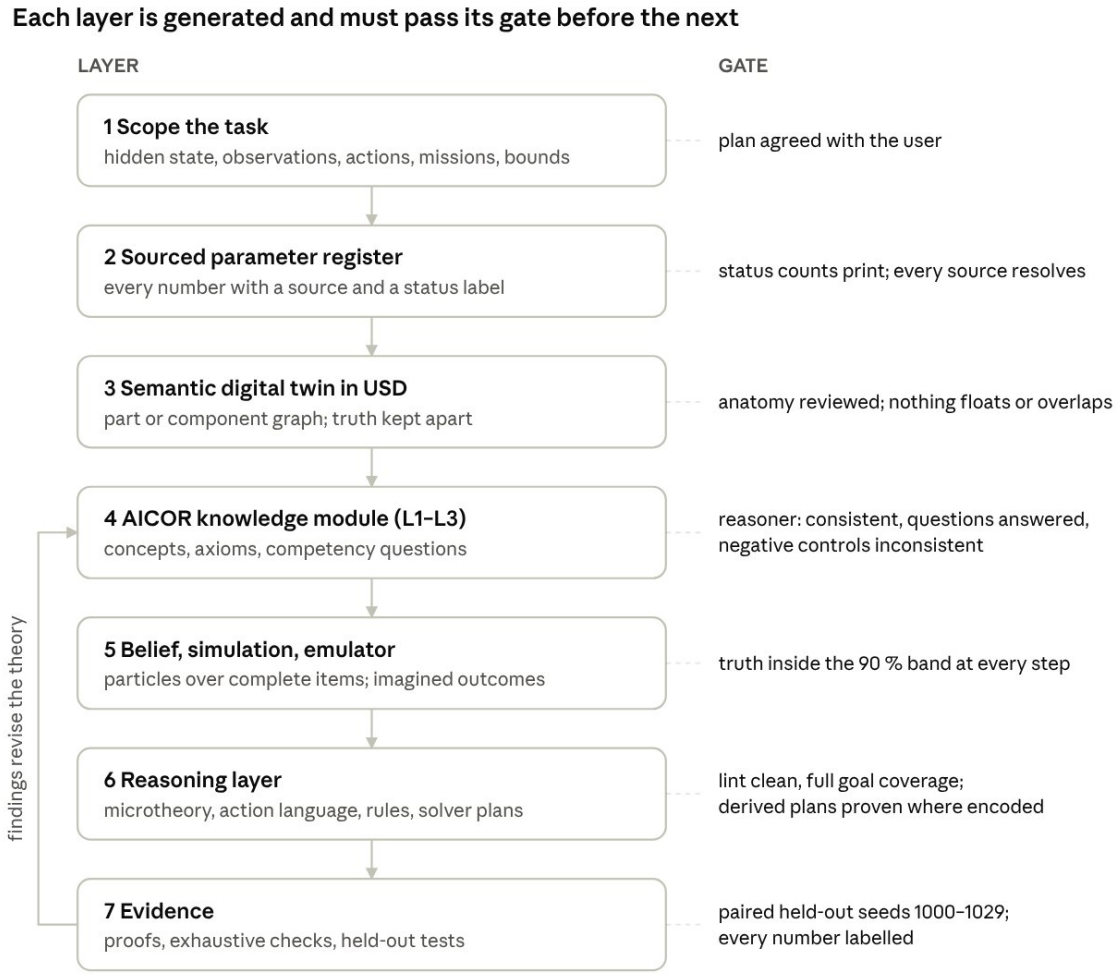
	Predicted	Observed	n
Retries, six belief-based conditions	0.95	**0.66**	103

The retry failure has a visible mechanism. After a cautious no-crack at 0.25 m/s, the re-decided retry jumps to 0.55 m/s with predicted P(success) 0.98 and shatters. Tempering the strike likelihood was not enough. For the avocado, the analogous calibration is also optimistic: Task B success predicted 0.996 against an observed 0.966 (n = 350, Brier 0.034).

12 Generating the representations and reasoning tools

The avocado and the egg were not modelled by hand from scratch. They were produced by an AI agent following generative procedures (packaged as skills, with scaffolders and checkers) that turn a domain into a sourced twin, a knowledge module and an executable reasoning layer. Each layer ends in a gate the agent must pass before going on.

The point for science and engineering is reproducibility of the representation itself. A new item, device or site goes through the same sequence of layers, checks and evidence as the reference implementation.



The agent builds from scope to evidence, and a layer counts only once its gate passes. Findings at the evidence stage flow back into the knowledge module as revised concepts, restricted scope

or retracted claims.

Three generators, one discipline.

Generator	Produces	Layers or phases	Example gates
Food SDT stack	An item twin from register to microtheory	L0–L11	anatomy plate reviewed by the user; truth inside the 90 % band at every belief step; held-out seeds 1000–1029 with ablations and misspecified plants
Isaac twin	A sourced twin of a site, cell or object, with ontology and CRAM agent	P0–P8	register prints status counts and every source resolves; no asset below the floor or floating; physics tests N/N; every top-level designator runs; the report rebuilds from clean; a hostile self-review
Conceptual CRAM activity	A research-activity layer: ontology module, action language, plan library, rules, adapter, evaluation	Step 0 plan, then P1–P9	lint without errors or META findings; 0 predicates missing from the ontology; no schema without a goal; nothing unrefinable; paired held-out evaluation

What is generated automatically.

- **Skeletons.** `ccram new` emits an activity that already lints clean, has complete goal coverage, passes its tests and runs one plan on a toy emulator. `scaffold_item.py` maps 25 files of an item to their avocado counterparts and writes stubs with the interfaces to implement.
- **Status reports.** These show which layers exist and which gates pass. For the egg, the register has 59 entries and L9–L10 are missing; for the banana, only anatomy exists.
- **The action language itself.**
  - The lexicon is derived from the ontology.
  - The designator grammar comes from the task axioms: required roles from `isTaskOf some R`, admissible roles from `only (…)`.
  - Designators translate to description-logic class expressions, so a reasoner can classify a planned action.
- **Everything downstream of the builders.** That covers USD assets, ABoxes, a USD sublayer that writes inferred ontology types back onto 122,931 scene nodes of the factory twin, work instructions, plans, reports and PDFs. Every result JSON stores the code and input hashes.

What the agent writes, and what the user decides. The agent writes:

- registers with sources and statuses;
- anatomy, layout and product models;
- TBoxes and axioms;
- solver encodings, adapters, rules, controllers and tests;
- self-reviews that list its own findings.

The user decides research questions, success criteria, norms, layout choices and the acceptance of anatomy. The procedure keeps that division explicit: engineering defaults the agent picks

are labelled ILLUSTRATIVE, and a question goes to the user only when it is a research question.

13 Scoping tasks

Generation starts with scoping: before any model is built, the procedure fixes what the task is, what world it lives in, what the robot can observe and what counts as success. The scope then travels with every claim, so the agent can answer ”out of scope” instead of guessing.

The item analysis. For a food item, step 0 asks:

- which hidden state decides the task;
- which observation channels reveal what;
- which transforming actions exist, with the forces, torques, temperatures and times that decide their effects;
- which time processes run, such as ripening, staling and browning;
- which missions must hold, and which risks must be prevented with which chance bounds.

The answer decides which layers carry weight. For a tomato that is slicing and bruising, for a mug fill level and temperature, for an egg freshness and the crack.

The research-activity plan does the same one level up. It fixes:

- the research question;
- the robot-world model (RWMS): objects, places, agents and a declared boundary;
- what is observable and what stays hidden truth;
- the actions with goal templates;
- the failures and closing events;
- the scenarios;
- the evaluation design with its baseline and its fallible condition.

Problems are stated outside the RWMS and assessed against it.

Scope is checked at run time.

- **The gate.** Eight AICOR admissibility conditions form the gate: grounded, feasible, mechanism-compatible, effect-expected, invariant-preserving, monitorable, has-fallback and in-scope. Closing events are refused there, never handled.
- **The commitment gate.** It answers *out-of-scope* when P(in scope) < 0.5, and *unsupported* when the concepts the request needs are not operationally covered.
- **Concepts.** A concept enters a theory only if it is perceivable, actionable and traceable.
- **Abstention.** With no known belonging place, the put-away agent asks instead of guessing.

Scope moves with evidence.

- When a sticky cultivar fell outside the avocado theory’s family (41/60), the scope was restricted and the claim retracted.

- In the lab twin, the experiment planner refused three requirements, each with its reason: an information gain of 0.70 bits that no candidate experiment reaches, a contaminated setup (safety rule S8) and an open door (S6).
- Each refusal is a statement about the boundary of the theory. The system can make it because the scope is represented, not implied.

14 Creating AICOR framework theories

Each generated twin comes with a theory in the AICOR framework. It is a knowledge module that extends the frozen AICOR hub ontology, is layered from bundle to logic, and feeds what it cannot express back to the framework as findings.

The module layers.

- **L1:** the knowledge-base bundle.
- **L2:** the conceptual framework, an OWL module with competency questions.
- **L3:** the logical axiomatisation, with axioms and derivations.

SDT nodes are kept distinct from the things they represent (node $\neq$ referent), and the module types both.

Module	Size	Extends
Lab automation (Cell 01)	412 concepts, 176 properties, 13 sub-modules (devices 73, site 63, kernel 53, tasks 35, experiment-to-motion 34)	AICOR hub, robot-scientist vertical, VRB devices, normative admissibility
Cuxhaven nacelle site	548 classes (155 defined) in 11 modules, 238 properties, 2,721 logical axioms, SROIQ(D)	AICOR hub, two Siemens verticals
Food items, activities	Item ontologies (egg 143 lines; booth 41 classes) aligned to SOMA/DUL	the ccram language ontology

Microtheories sit inside the modules.

- **Floss machine:** quantities, five modes, timings with statuses and two closing events (pouring into the moving head, contact with the spinning head).
- **Egg shell:** modes intact, microcracked, cracked, opened and shattered, where shattered signals a failure.
- **Avocado:** a versioned package $T = \langle \Sigma, A, M, C, E \rangle$ with held-out claims.

In the lab, the L3 theory is a chain of nine axioms (E2M1–E2M9). It runs:

1. from a required information gain;
2. through the EIG-optimal experiment;
3. to material, scene and motion designators;
4. under safety rules S1–S9.

Theories are revised, not frozen.

- The avocado's learning path (section 8) admits concepts when a request exposes a gap, and retracts a claim when evidence refutes it.

- The lab module's deferred register records 12 places where the framework itself fell short, as proposals back to AICOR. Examples:
  - there is no vocabulary for non-robot mechanisms and sensors;
  - the OWL exporter reads `EquivalentTo` as a necessary condition only, so 7 definitions lose their sufficient half;
  - an `exactly 96 Well` cardinality makes HermiT intractable;
  - three SOMA classes are unsatisfiable upstream.

Generation therefore also tests the framework it builds on.

15 Correctness: what is proven, what is checked, what is only tested

Because the representations are generated together with their reasoning tools, correctness obligations can be generated with them, and some are discharged by proof.

The proofs we have are real but scoped:

- entailment in the ontologies;
- optimality of a derived experiment plan;
- safety, timing and gain properties of that plan for all unknown initial states;
- a refinement theorem from the plan down to robot motions.

Everything below that level is checked mechanically or tested empirically, and the article keeps the three apart.

Kind	What is established	Result	Scope limit
Proof: OWL DL reasoning (HermiT, Konclude)	Consistency, coherence and classification of each module	Lab: 1,284 classes, consistent, 0 unsatisfiable. Cuxhaven: 1,384 classes consistent; 216,928 individuals realised. Booth: the pour designator is a prohibited action exactly when the machine is producing	Holds for the axioms as exported: 41 non-normative restrictions skipped; `EquivalentTo` sufficiency lost. Open world: "not provably prohibited" is not "safe"
Proof: competency questions by entailment	The ontology answers its design questions; negative controls are inconsistent as required	Lab: 18/18 TBox CQs, 6/6 negative controls. Cuxhaven: 9/9 TBox and 22/22 ABox CQs, 23/23 negative controls (e.g. crane capacity below load, 7 stator segments instead of 6)	ABox answers compared with independently computed expectations
Proof: answer-set optimality (clingo)	The chosen experiment is least-cost, then highest-EIG, among 45 candidates; refusals are proofs of infeasibility	For a 0.5-bit requirement: EIG 0.544, optimal; 3 requirements refused with reasons	Within the finite candidate frame and apportioned durations

Kind	What is established	Result	Scope limit
Proof: SMT symbolic execution (z3)	V1 executability (278 preconditions), V2 safety (115 checks), V3 time window, V4 realisation of doses, images and arms, V5 gain and least cost	All 7 theorems hold; 10 mutant plans each caught	For every unknown initial tool and stage state, durations within ±20 %
Proof: refinement theorem	Every execution of the 151-step plan (111 motions) realises the plan, or a monitor stops it into a handler whose guarantee holds	Obligations O1–O7 discharged; 8 negative controls caught	Relative to assumptions A1–A6 (statechart semantics, monitor soundness, pose tolerance, grasp holds, device contracts, reachable frames). Not proved: progress, physics of insertion, biology
Exhaustive check	Structure and coverage	Product model: 12 structural checks, 0 issues over 49 components and 69 bolted joints. Activity lint and goal coverage: 0 gaps. Kernel vs planner: 151/151 steps identical. Audit trails: tampering detected in every scenario	Exhaustive for what the checker encodes
Empirical test	Behaviour in simulation	Device physics tests N/N (e.g. lab station H4 14/14, imager L17 15/15). Collisions: 0 pairs over 28 factory clips, down from 1,109 in round 1. Held-out evaluations of sections 8 and 11	Collisions sampled every 0.5 s, not swept continuously; shared plant and robot laws

The proofs found real errors.

- z3 rejected the first encoding's nominal-duration time window, which is where the ±20 % margin came from.
- z3 refuted a derived plan that ran quality control after the plate was already on the imager. The old plan fails at step 24, and all seven plans were re-derived.
- HermiT found six unsatisfiable kernel categories, where subclassing had been confused with parthood.

The proofs do not close the gap to the world.

- The plan's expected information gain is proven to be at least 0.5 bits (0.544). On the twin's culture model, the realised gain averaged 0.21 bits over 40 runs, because two of the three candidate models cannot be told apart. That is a proof about the plan's model, and an empirical finding about the world it plans for.
- The axiom audit's 22 counterexamples (section 8) mark where the symbolic theory needs hidden attributes before such proofs can extend to food.

16 Beyond food: devices, tools, meals, laboratories, apartments and factories

Food is the example, not the scope. The same procedure has produced twins and reasoning layers for a machine, laboratory devices, tools, meals, a laboratory cell, an apartment kitchen and a factory site. Each has a sourced register, a part or component graph in USD, an ontology module and a reasoning tool fitted to what decides its tasks.

Kind	Twin	What decides the task	Reasoning tool	Evidence (status)
Food items	Avocado and egg, full stack; 14 more items (apple to pepper) at anatomy and register	Hidden ripeness, seed, shell strength, age	Particle belief, imagined items, chance-bounded decisions, microtheory	Sections 4–11 (MEASURED in simulation)
Meals	Guacamole missions; eggs into a batter	Timing, browning, freshness of each ingredient	Lifetime models with chance bounds; rule chain from float test to the emptying gate	3 missions decided (1/4, 3/24, 2/12 feasible plans); a stale egg refused at the gate
Tools	Knife, spoon and strike surfaces; a 5-tool magazine in the lab; a nutrunner in the factory	Which tool may act on what, with what force or torque	Axioms with `UnsafeToolUse`; belief-chosen strike surface; governance checks	Knife strike refused at hazard P 1.00; an M48 bolt needs 4,929 N m against a 2,200 N m tool limit, so that station design is rejected
Machines and devices	Cotton candy floss machine; lab quarantine station, plate handler, incubator, liquid handler, imager	Device modes and the actions each mode permits	Process microtheory with closing events; device contracts and controllers	Pouring is classified as prohibited exactly when the head is producing; 62 machine tests pass; device physics tests 10/10 to 15/15
Laboratories	Cell 01 (robot scientist): stations, three robots, a culture model	Which experiment yields the required information, and how to execute it safely	Ontology module (412 concepts), experiment-to-motion derivation, z3 proofs, kernel	Plans proven (section 15); twin execution 151/151 steps; realised gain 0.21 vs 0.54 bits expected
Apartments	PR2 apartment kitchen: putting things where they belong	Norms of belonging, requests, what a person may overrule	Norms as rules, belonging verdicts, a gate against dirty dishes into the cupboard, abstention	30 kitchens: verdict accuracy 1.0 with perfect survey, 0.973 with fallible survey; 0 over-tidy; 25/30 tasks succeed
Factories	Cuxhaven nacelle site: hall, 3 lines, cranes, vehicles, 29 station steps, quay	Product structure, station sequence, crane and vehicle capacity, work content	Ontology module (548 classes), product model checks, line simulation, governance constraints, collision checking	0 issues across 49 components and 69 bolted joints; 587 work tasks, 16 % robot-ready (ILLUSTRATIVE ratings); 0 collisions over 28 clips

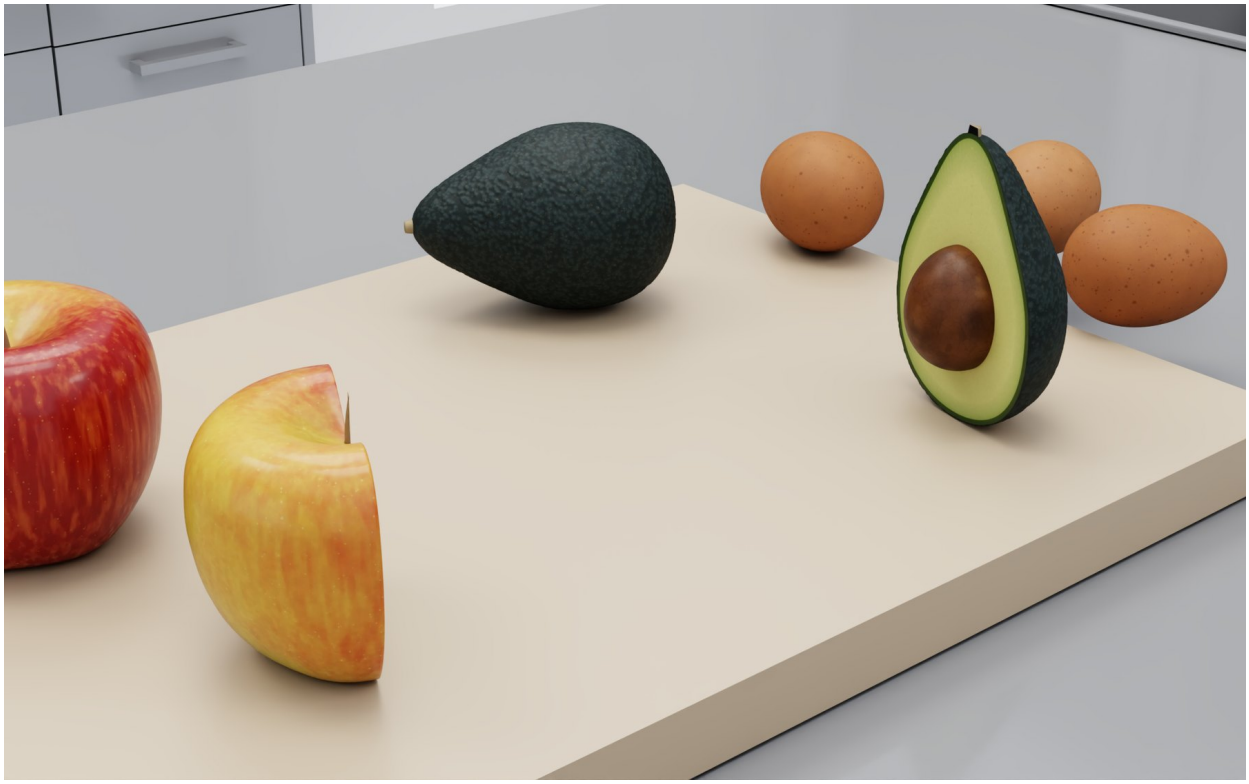


Figure 11: Apartment: food twins on the kitchen island of the PR2 apartment.



Figure 12: Laboratory: Cell 01, where experiments are derived from a required information gain and executed as proven plans.

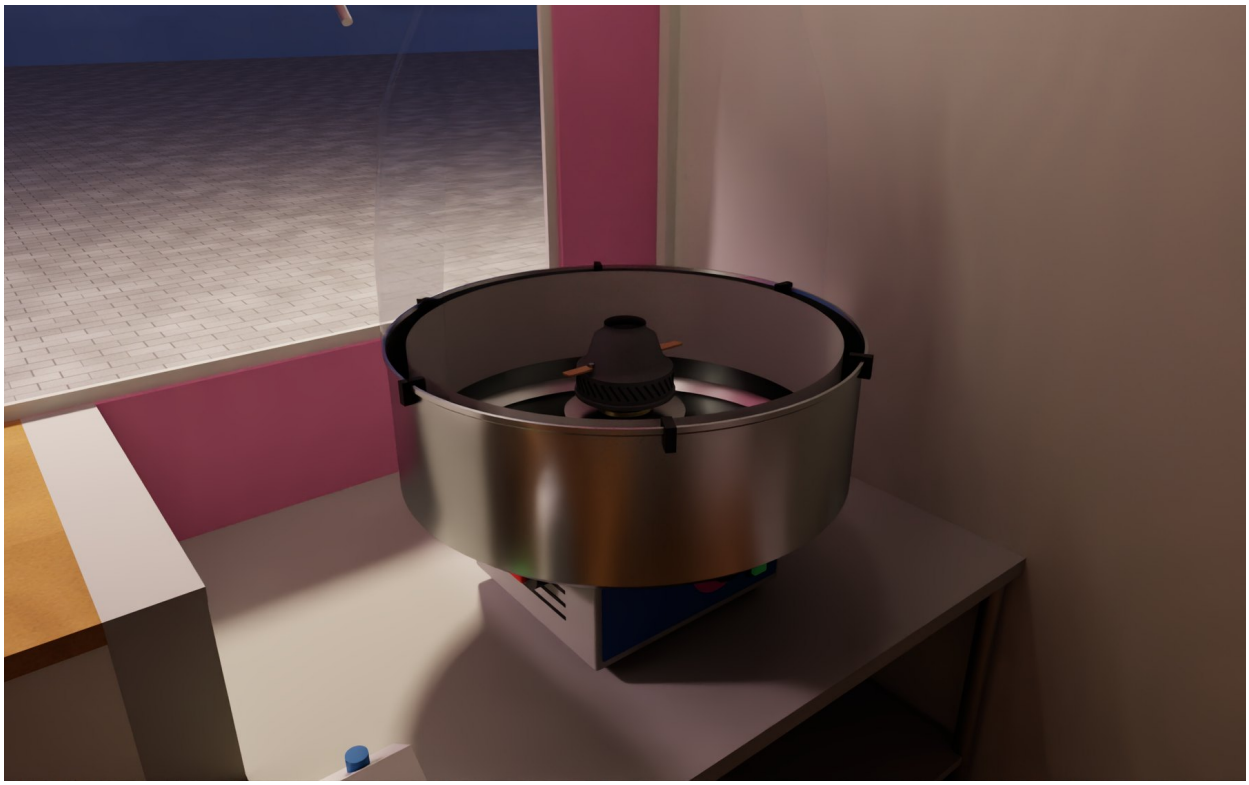


Figure 13: Machine: the floss machine, whose modes decide which actions are prohibited.

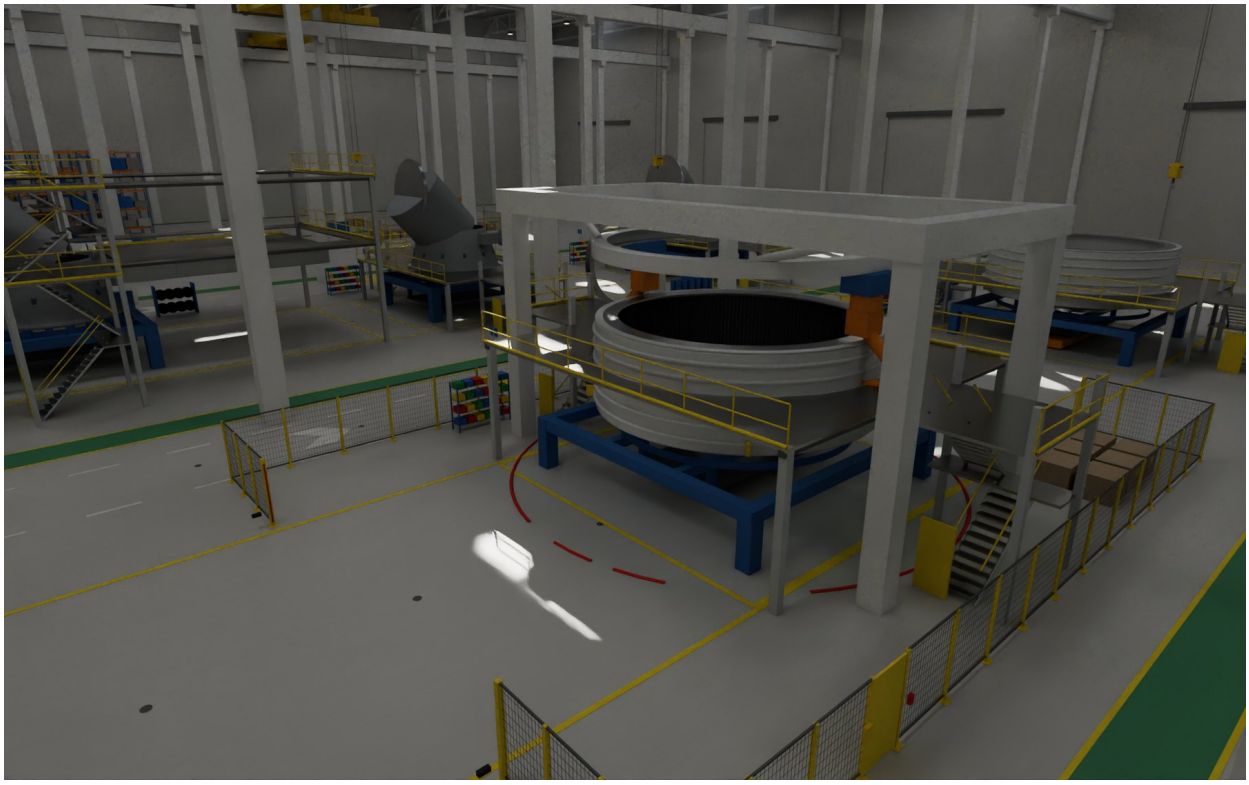


Figure 14: Factory: the magnet-insertion station of the Cuxhaven nacelle site, one of 29 station steps.

What transfers is the analysis, not the content. Every twin answers the same questions:

- which hidden state decides the task;
- how it is observed;
- which actions transform it, and under which guards;
- which processes run in time;
- which goals and risks must be bounded.

What changes is which layer carries the weight:

- belief and simulation for food;
- modes and closing events for a machine;
- solver-checked plans for a laboratory;
- norms for an apartment;
- product structure, capacity and collision for a factory.

The registers keep every case honest about its evidence. The factory's 104 parameters are 42 SOURCED, 33 ILLUSTRATIVE, 10 from OpenStreetMap and 0 GAP; the floss machine's 133 are 58 SOURCED and 28 ILLUSTRATIVE.

17 Connections to cognitive AI

The twins can be read as an engineering answer to a programme from computational cognitive science. Robots should hold causal, generative models of the hidden structure that drives their tasks, run them as approximate probabilistic simulations, and connect them to language. Four lines of work make this concrete. They sharpen what we can claim, and they point to what is still missing.

- **Dark, Beyond Deep** (Zhu et al. 2020) argues that what drives scenes and tasks is invisible in pixels. It calls this "dark matter" and names five domains: functionality, physics, intent, causality and utility (FPICU). It also calls for "small data for big tasks" instead of big data for narrow ones.
- **The game engine in the head** (Battaglia, Hamrick and Tenenbaum 2013; Ullman, Spelke, Battaglia and Tenenbaum 2017) proposes that people judge physical scenes with an intuitive physics engine. This engine runs approximate, probabilistic simulations over uncertain scenes and, like a video-game engine, takes shortcuts to run in real time.
- **Building Machines That Learn and Think Like People** (Lake, Ullman, Tenenbaum and Gershman 2017) asks machines to:
  - build causal models rather than recognise patterns;
  - ground learning in intuitive physics and psychology;
  - use compositionality and learning-to-learn to generalise from little data.
- **From Word Models to World Models** (Wong et al. 2023) proposes rational meaning construction. A language model translates language into a probabilistic language of thought, and Bayesian inference over the resulting generative world model does the thinking. The authors argue that systems should be architected around thought, with language models as one plug-in among others.

How the twins realise these ideas, and what is still open:

Idea	From	How the twins realise it	Still open
Dark entities decide tasks	Zhu et al.	Hidden state as SDT variables: ripeness, testa side, shell strength, age. Observability classes grade how dark each part is	Intent appears only as norms and requests (put-away)
Fluents: time-varying states perceived through causal change	Zhu et al.	State predicates are fluents that hold at points of a trajectory; goals are templates over state evolutions	—
Functionality and affordance	Zhu et al.	Dispositions in the ontology (graspability, containment), strike surfaces, belonging places, unsafe tool use	Affordances are asserted, not learned from use
Utility drives decisions	Zhu et al.	Missions with chance bounds, expected quality, value of information	Utilities are ILLUSTRATIVE, not elicited from people
An intuitive physics engine: approximate probabilistic simulation	Battaglia et al.; Ullman et al.	Imagined items: mechanics run on particles drawn from the belief, about 1 ms each. They yield P(success), motion limits and expectations for monitoring	No systematic study of where the fast models fail
Engines cut corners to run in real time	Ullman et al.	Two tiers: Isaac FEM and particle physics run offline and calibrate the reduced-order models the robot runs online. Example: the FEM squeeze table becomes the stiffness model	Error between the tiers is not yet reported per model
Surprise as a signal	Ullman et al. (violation of expectation in infants)	Monitoring places each measurement in the imagined distribution: expected, unexpected or surprising	Thresholds are ILLUSTRATIVE
Causal models, not pattern recognition	Lake et al.	Interval axioms with guarded branches; held-out claims; ablations that name the concept carrying a result	22 counterexamples show attributes the relational layer lacks
Compositionality	Lake et al.	Partonomies in USD, a factory product model of 49 components, designators composed from ontology terms	—
Model building from sparse data	Lake et al.	Theory revision T0–T5: concepts admitted, scope restricted, a claim retracted	Revisions are proposed by scripted agents

Idea	From	How the twins realise it	Still open
Learning-to-learn	Lake et al.	Generators reuse the avocado stack as the template for new items, devices and sites	The building agent learns to learn; the robot does not yet
Language into a probabilistic language of thought	Wong et al.	Designators form a language of action derived from the ontology; the SDT belief is a generative model the robot queries	Natural-language requests are not yet translated into goals or observations
Architect around thought, with language models as plug-ins	Wong et al.	A language-model agent builds the world models and reasoning tools; the robot decides on the models, not on the language model	—

Where this work differs.

- **What the models are for.** The cognitive models explain human judgments. Ours must also act: the same simulation that predicts an outcome sets a force limit, chooses a strike speed and judges the result.
- **Scale.** Our world models are built per item, device and site from sourced registers, with a status on every number. Their scale comes from automated generation, not from hand-crafted toy domains.
- **Kinds of guarantee.** Next to probabilistic inference stand reasoner entailments, solver-proven plans and safety rules with bound 0 that no utility can buy (section 15).

What the four lines suggest next.

1. **Language as observation.** Apply rational meaning construction to the SDT node. ”This one feels soft” or ”the eggs are from last month” become likelihood terms on the belief. ”Leave the coffee out until I’m done” becomes a goal template with its obligation.
2. **A physics-illusion study for the robot.** Measure where the fast mental engine and the slow faithful engine disagree, per process model, the way the human illusions were mapped. The darkened skin that misled the ripeness belief is a first case.
3. **Measure darkness.** Compute, per variable, how much each observation channel narrows the belief, as in the table of section 4. That turns FPICU’s ”darkness” into a quantity that value-of-information decisions can use.
4. **Intuitive psychology in the kitchen.** Infer who wants what, and when, by inverse planning over people’s actions, instead of encoding norms and requests only.
5. **Amortised inference under theory authority.** Learn fast proposals for the particle filter and the imagined items, as Lake et al. suggest for structured models, while the theory and its safety rules keep the final say.

18 Limits and next steps

All of this is simulation evidence: nothing has yet been measured on a real avocado, a real egg or a real robot. The plant and the robot share the same laws, so the results show internal competence, not physical adequacy.

What is emulated or simplified.

- There is no contact physics for knife, twist or strike. Fingers, knife and spoon pass through the fruit.
- The PR2 runs are kinematic, and the egg kitchen is a hybrid: CRAM motions around emulator outcomes.
- 46 % of the avocado register and 24 of the egg's 59 entries are ILLUSTRATIVE. So are most decision thresholds: chance bounds, VOI thresholds, the caution factor and the tempering.
- Perception is validated on renders only. The photo ripeness likelihood is not calibrated, and the renderer's dark skin misled the PR2 run (3.97 believed vs 2.96 true).

What the results say is weak.

- In both foods, predicted success is optimistic: avocado 0.996 vs 0.966; egg retries 0.95 vs 0.66.
- Misspecification dominates the chance bound. 9 of 30 eggs shattered under "P(shatter) $\leq$ 0.05", and weak avocado flesh (bruise threshold × 0.6) leaves 4/30 correct.
- Several differences are not significant at n = 30: belief vs fixed strike (p = 0.125), caution (p = 0.15), and feedback in Task A (p = 0.62).
- Photo-only matches the belief band on well-specified plants. This is because the "photo" is a symbolic reading of the plant's own colour law.
- 22 axiom counterexamples show the relational theory does not determine outcomes without hidden attributes.
- The executed "half now, half tomorrow" mission in the live kitchen missed flesh intact and edible at service.

Next steps.

- **Calibrate retries:** model the information in a no-crack properly rather than tempering it, and re-run the paired strike evaluation.
- **Make the theory set-valued:** let transitions depend on hidden attributes and test against the 22 counterexamples.
- **Ground physically:** contact physics for knife and strike, with real-image perception on recorded fruit and eggs.
- **Belief over kind and state with probe-first** (milestone M4): carry the egg's belief-decided servants over to knowledge-heavy activities such as putting things where they belong.
- **Settle `:violates`:** decide whether a failure's declared goal parts are the right semantics for accepted deviations.

Code inconsistencies found while preparing this article:

- The plant's clean-release law uses width 0.25; the planner and the learning check use 0.15.

- The schema's squeeze guard says 0.5 × penetration pressure; the register says 0.6.

Neither is reflected in the numbers above yet.

Sources

Every number above was read from these files on 2026-10-22. Paths are relative to the project repositories.

Topic	Source
Avocado SDT, overview and "not yet" lists	`HouseholdTwin/docs/AVOCADO_SDT.md`
Parameter registers (avocado 48, egg 59 entries)	`HouseholdTwin/food/params.py`
Part graph and truth/belief layers	`HouseholdTwin/food/avocado.py`, `food/usd_partgraph.py`, `assets/food/avocado_4/`
Avocado belief and convergence	`food/avocado_belief.py`, `assets/food/avocado_4/belief/timeline.json`
Imagined items, monitoring, VOI	`HouseholdTwin/cram/avocado_sdt.py`, `cram/results/pr2_run.json`
Chance-constrained agent	`cram/avocado_agent.py`, `report/cache/avocado_agent_missions.json`
Microtheory, audit, learning, RL	`HouseholdTwin/theory/`, `theory/results/{microtheory,axiom_audit,learning,prl}.json`, `report/avocado_microtheory.tex`
Egg model and belief	`food/egg.py`, `food/egg_crack.py`, `food/egg_belief.py`, `test_reports/egg_belief_calibration.json`
Egg activity (ontology, library, rules, adapter)	`HouseholdTwin/activities/egg-cracking/`
Strike evaluation	`activities/egg-cracking/experiments/strike_eval.json`
Conceptual CRAM layer (kernel, language, rules)	`ConceptualCRAM/ccram/kernel.py`, `lang.py`, `rules.py`, `ontology/ccram-lang.ttl`
Design decisions D1–D10, Q2	Conceptual CRAM concept page
Generators and their gates	skills food-sdt-stack, isaac-twin, ccram-research-activity; ConceptualCRAM/ccram/generator.py, lang.py
Lab module, plan derivation and proofs	LabAutomationTwin/okf/lab-automation-twin (reasoner and CQ reports, DEFERRED-REGISTER.md, derivation/out/proof_g050.json), docs/rsa/REFINEMENT-PROOF.md, test_reports/
Factory site module, product model, checks	CuxhavenSiteTwin/okf/cuxhaven-site, product/catalog.py, data/work_instructions.json, data/ng_governance.json, data/collisions_next7/summary.json, site/params.py
Floss machine and booth	CottonCandyTwin/cram/concept/booth.lisp, ontology/reasoning_demo.json, test_reports/cram_pytest.xml
Apartment put-away	HouseholdTwin/activities/put-away/experiments/guard_eval.json

Topic	Source
Cognitive AI (section 17)	Zhu et al. 2020, Dark, Beyond Deep, arXiv:2004.09044 · Battaglia, Hamrick, Tenenbaum 2013, PNAS 110(45):18327–18332 · Ullman, Spelke, Battaglia, Tenenbaum 2017, Trends Cogn. Sci. 21(9):649–665 · Lake, Ullman, Tenenbaum, Gershman 2017, Building Machines That Learn and Think Like People, Behav. Brain Sci., arXiv:1604.00289 · Wong et al. 2023, From Word Models to World Models, arXiv:2306.12672